LAB 7

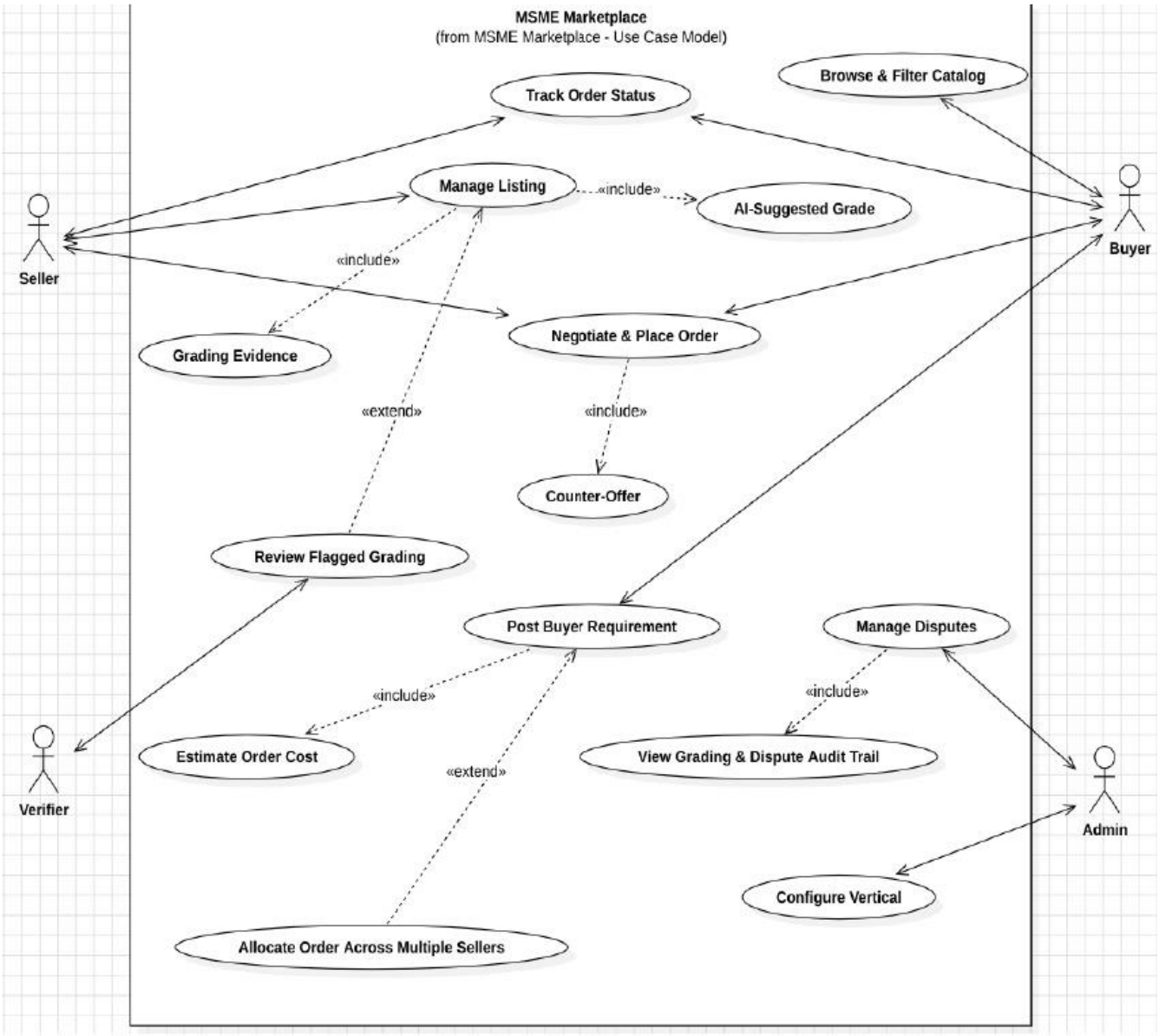
AIM - For 3 use cases of your project, draw the appropriate activity diagrams.

TEAM MEMBERS -

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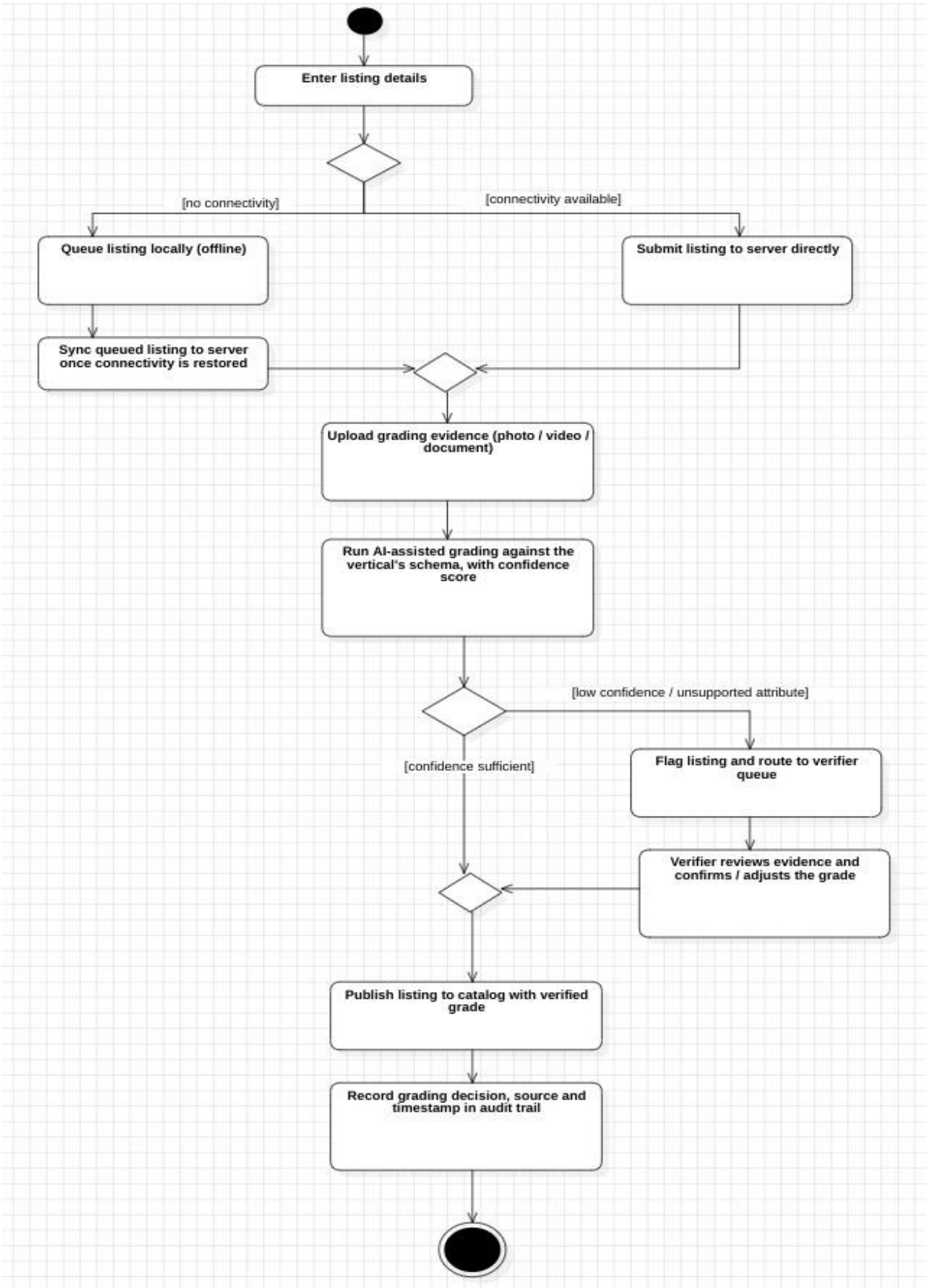
USE CASE DIAGRAM -



ACTIVITY DIAGRAMS:

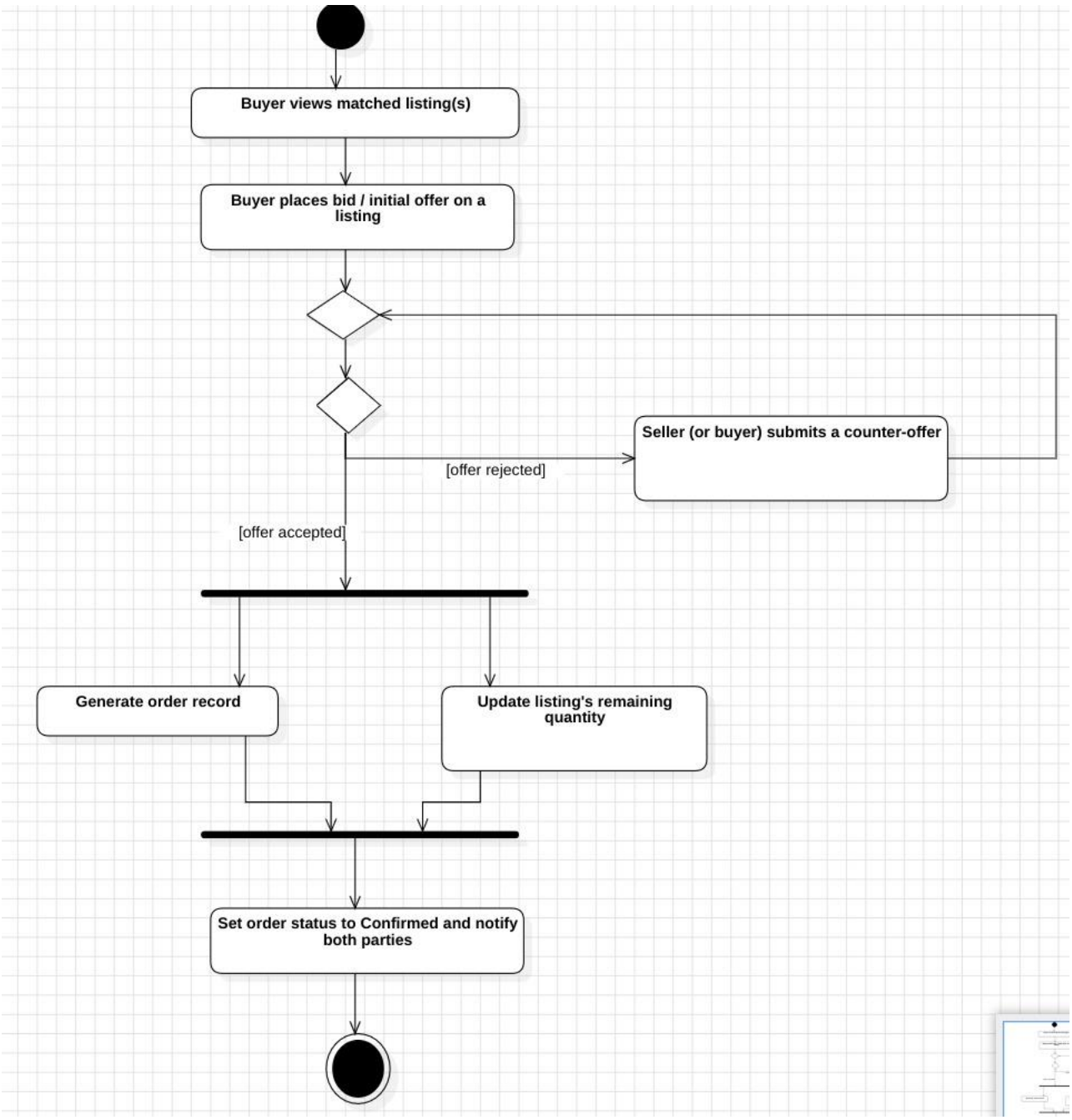
1. Manage Listing

- Seller starts by entering listing details (commodity, quantity, unit, location, etc.).
- System checks if there is internet connectivity.
    - If **no connectivity**, the listing is queued locally and synced later when connection is restored.
    - If **connectivity is available**, the listing is submitted directly to the server.
- Both paths then join together.
- Seller uploads grading evidence (photo/video/document).
- System runs AI-assisted grading and generates a confidence score.
- System checks the confidence score.
    - If **confidence is sufficient**, it proceeds directly.
    - If **confidence is low**, the listing is flagged and sent to a verifier, who manually reviews and adjusts the grade.
- Both paths join again.
- Listing is published to the catalog with the verified grade.
- The grading decision, source, and timestamp are recorded in the audit trail.
- Process ends.


Enter listing details
[no connectivity]
[connectivity available]
Queue listing locally (offline)
Submit listing to server directly
Sync queued listing to server once connectivity is restored
Upload grading evidence (photo / video / document)
Run AI-assisted grading against the vertical's schema, with confidence score
[low confidence / unsupported attribute]
[confidence sufficient]
Flag listing and route to verifier queue
Verifier reviews evidence and confirms / adjusts the grade
Publish listing to catalog with verified grade
Record grading decision, source and timestamp in audit trail

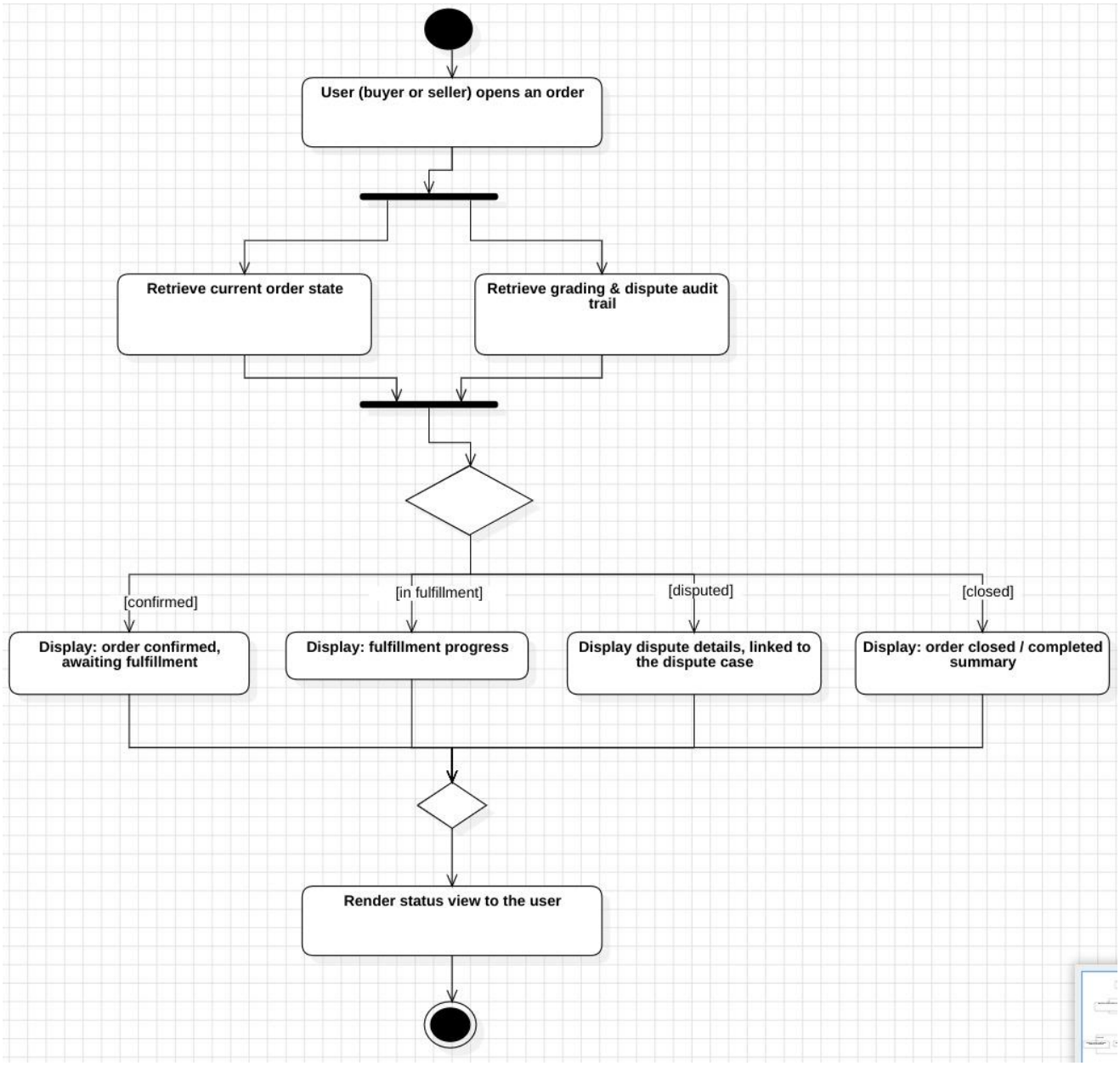
2. Negotiate & Place Order

- Buyer starts by viewing matched listings.
- Buyer places a bid/initial offer on a listing.
- System checks if the offer is accepted.
    - If **rejected**, the seller (or buyer) submits a counter-offer, and the process loops back to check acceptance again.
    - If **accepted**, the process moves forward.
- At this point, the flow **forks** into two parallel activities:
    - Generate the order record.
    - Update the listing's remaining quantity.
- Both activities **join** back together once completed.
- Order status is set to "Confirmed" and both buyer and seller are notified.
- Process ends.


Buyer views matched listing(s)
Buyer places bid / initial offer on a listing
Seller (or buyer) submits a counter-offer
[offer rejected]
[offer accepted]
Generate order record
Update listing's remaining quantity
Set order status to Confirmed and notify both parties

3. Track Order Status

- User (buyer or seller) opens an order.
- The flow **forks** into two parallel activities:
    - Retrieve the current order state.
    - Retrieve the grading & dispute audit trail.
- Both activities **join** back together once completed.
- System checks the order's current state and branches into one of four outcomes:
    - **Confirmed** → shows order confirmed, awaiting fulfillment.
    - **In fulfillment** → shows fulfillment progress.
    - **Disputed** → shows dispute details linked to the case.
    - **Closed** → shows order closed/completed summary.
- All four outcomes merge back together.
- The final status is rendered/displayed to the user.
- Process ends.


User (buyer or seller) opens an order
Retrieve current order state
Retrieve grading & dispute audit trail
[confirmed]
[in fulfillment]
[disputed]
[closed]
Display: order confirmed, awaiting fulfillment
Display: fulfillment progress
Display dispute details, linked to the dispute case
Display: order closed / completed summary
Render status view to the user